

Support Team Performance

Scrum ceremonies

Sprint planning

- Team and product owner plan work for the current sprint.
- Scrum master or senior scrum master facilitates the planning session.

Daily scrum

- Team holds a daily stand-up meeting.
- Team members discuss work, request assistance, and assess progress toward the goal.
- Not a status meeting

Sprint review (can include demo)

- Held at the end of each sprint
- Team, product owner, stakeholders, or customers review progress and give feedback.

Sprint retrospective

- Team identifies and plans improvements in performance and collaboration.

Agile ceremonies

Product strategy meeting

- Product owner shares the product vision.

Daily stand-up or stand-up

- Team conducts a status meeting.
- Meeting is timeboxed, limited to 5 to 15 minutes.
- Not necessarily daily

Backlog refinement

- Product owner prioritizes items on backlog.

Project retrospective

- Team reviews work and processes at the end of a project.
- Similar to lessons learned

Scope validation: Process of customer acceptance of completed deliverables



Predictive project life cycle

- Acceptance criteria



Adaptive project life cycle

- Definition of Ready (DoR)
- Definition of Done (DoD)
- Acceptance criteria
- Iteration reviews



Hybrid project life cycle

- Any combination of the above

Evaluate and Manage Quality

Predictive projects

- Team members dislike and avoid work.
- Team members avoid increased responsibility.
- Team members need direction.

Adaptive projects

- Team, customer, and product owner set and meet the quality goal and metrics.
- Feedback from iterations monitor quality continuously
- Quality performance is measured with:
 - Service Level Agreements (SLAs)
 - KPIs
 - Contractual measures
 - Quality methods or frameworks, such as Lean Six Sigma

Quality Control Tools

Data Gathering Tools

- Checklists or check sheets
- Statistical sampling
- Questionnaires
- Surveys

Data Analysis Tools

- Performance reviews
- Root cause analysis

Data Representation Tools

- Cause-and-effect diagrams
- Scatter diagrams
- Control charts
- Histograms
- Pareto charts

Tailor performance measurement to the project context and stakeholders:

Scope

- Percentage of work completed
- Change requests

Schedule

- Actual duration of work against projected start and finish dates

Budget

- Actual costs
- Procurements vs. needs

Resources

- Team allocations, availability, or procurement
- Performance appraisals of team, including vendors
- Contract management

Quality

- Technical performance
- Defects

Performance reports

Milestone schedule

High-level visualization of progress on work against planned dates

Quality reports

Charts and reports based on the quality metrics collected

Earned value management (EVM) reports

Graphs and values based on EVM equations

Variance analysis reports

Graphs and their analysis comparing actual results to expected results

Work performance reports

Physical or electronic representation of work performance information compiled in project documents, intended to generate decisions, actions, or awareness

Dashboards

Physical or electronic progress summaries, usually with visuals or graphics to represent the larger dataset

EVM Measures for Cost Control

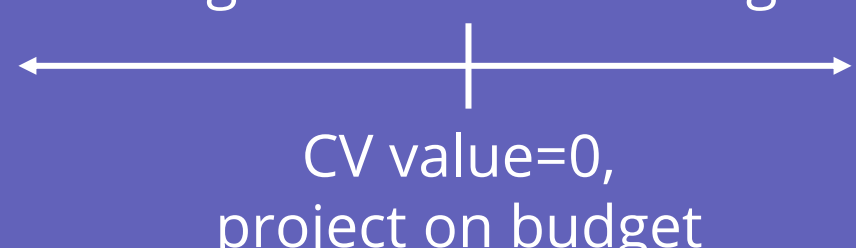
Is the project on budget?

Calculate **cost variance (CV)** to find the current amount of budget deficit or surplus.

$$CV = EV - AC$$

Project over budget

Project under budget



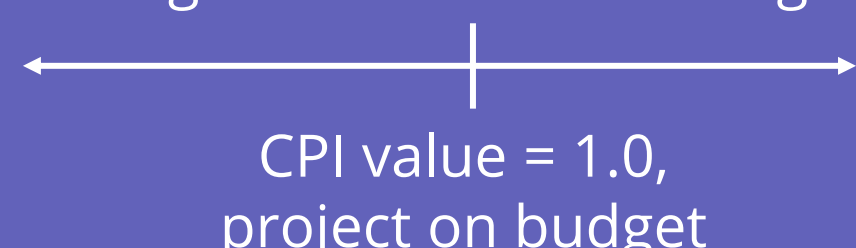
How efficient is your project?

Calculate **cost performance index (CPI)** to measure the cost efficiency of budgeted resources.

$$CPI = \frac{EV}{AC}$$

Project over budget

Project under budget



EVM Measures for Schedule Control

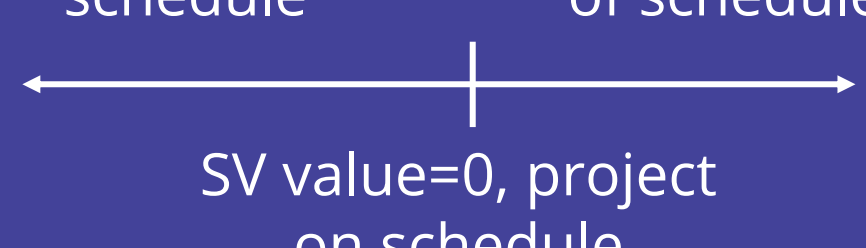
Is the project progressing on schedule?

Schedule variance measures performance by calculating the difference between EV and PV

$$SV = EV - PV$$

Project behind schedule

Project ahead of schedule



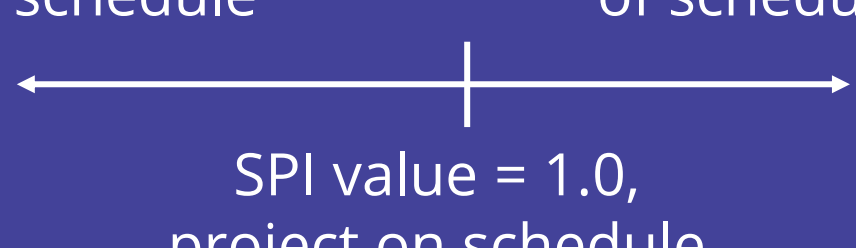
How efficiently is the team working?

Schedule performance index measures efficiency by calculating the ratio of EV to PV

$$SPI = \frac{EV}{PV}$$

Project behind schedule

Project ahead of schedule



What will be the project cost in total?

Use Estimate At Completion (EAC)

Based on:

- CPI: Current spending efficiency
- BAC: Budget at completion

$$EAC = \frac{BAC}{CPI}$$

How much more cost is required to complete the remainder of the project?

Use Estimate To Complete (ETC)

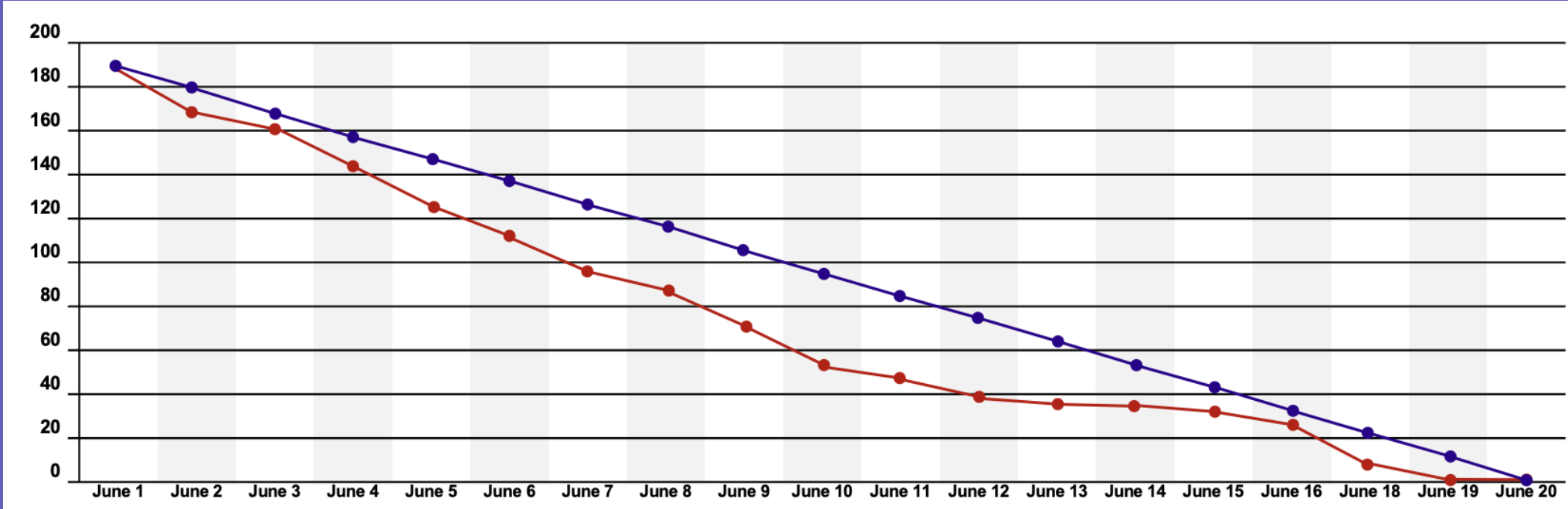
Based on:

- CPI
- AC (actual cost)

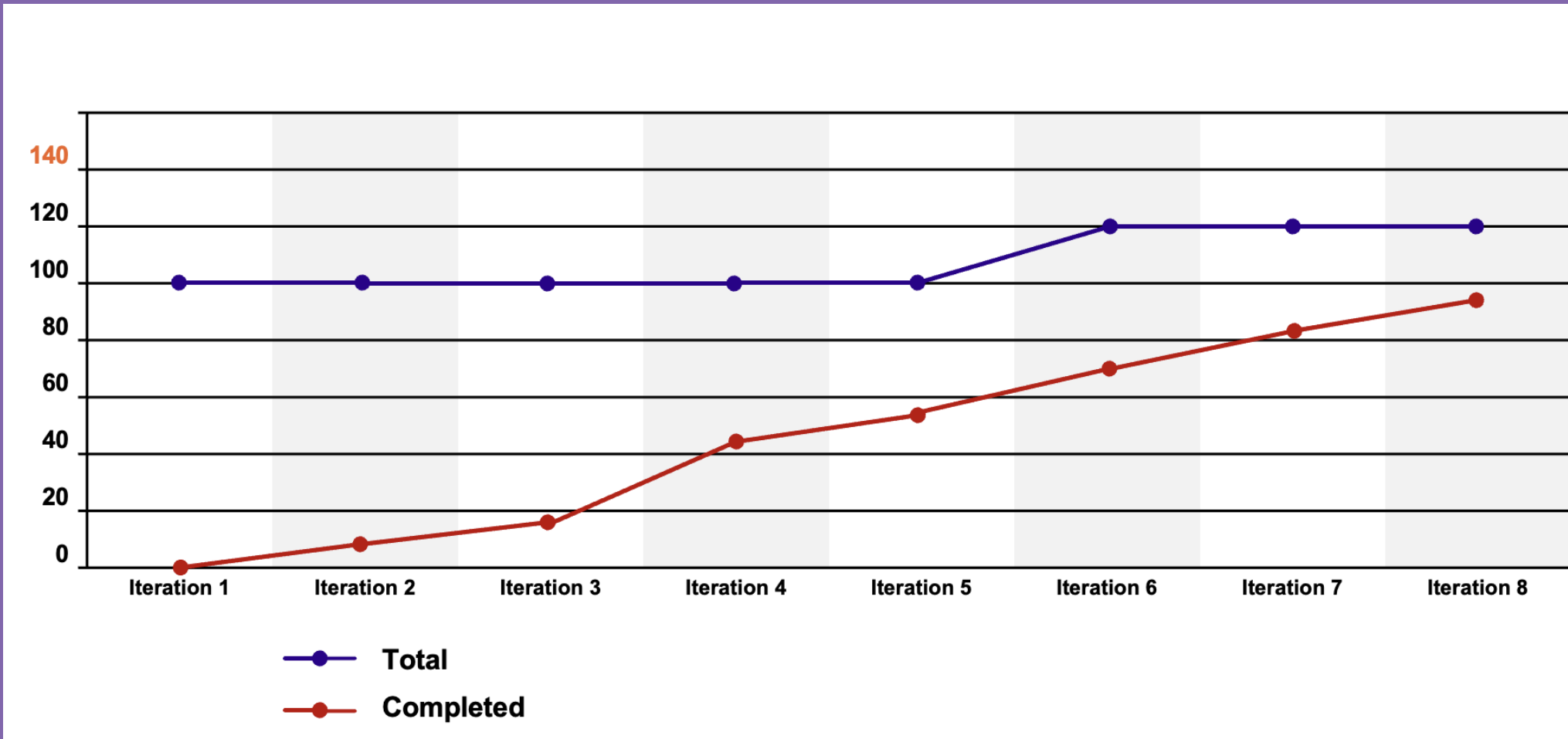
$$ETC = EAC - AC$$

Charts and graphs are visual representations of the team's progress.

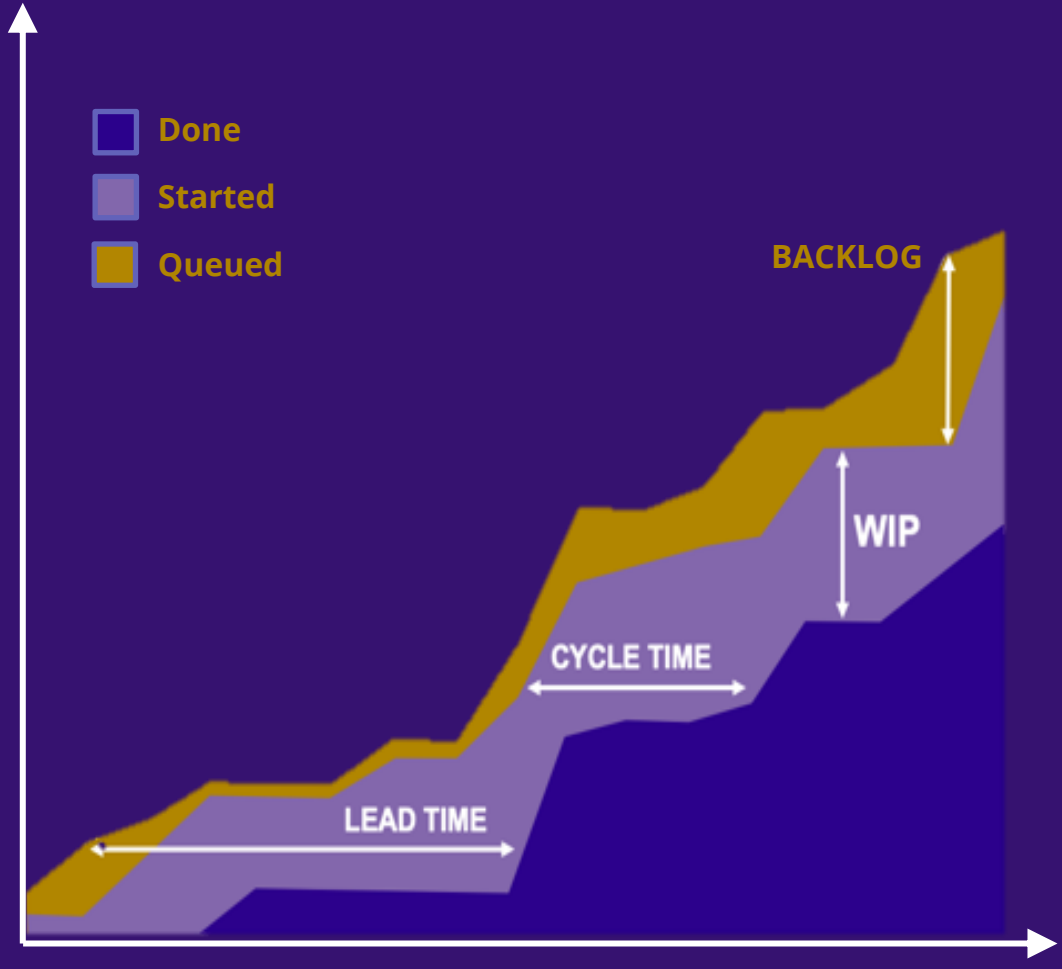
Burn-down chart (Iteration) is used to track the work to be completed within an iteration.



The **burn-up chart (Release)** shows the accumulated progress of completed work and is updated after each iteration.



The **continuous flow approach** measures the following:



- **WIP:** Measure of work in progress but not completed
- **Lead time:** Length of time work item goes through the entire process
- **Cycle time:** Length of time work item is being worked on
- **Throughput:** Number of items entering or exiting the system

The Cumulative Flow Diagram

Change Management Process Flowchart

